

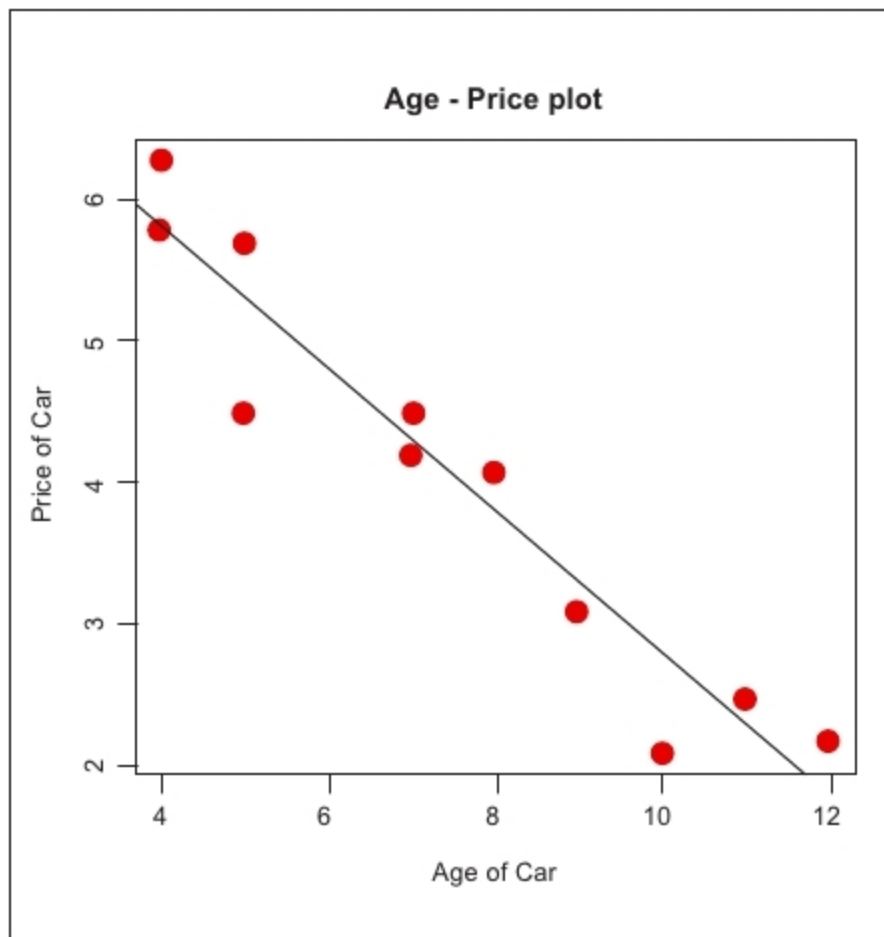


Figure 2: Car's age versus a car's price scatter plot and regression line

Graphical analysis

The results of the above analysis can be studied with R or Python. Here, for better understanding, I have done the same exercise with R. The age and price attributes are assigned to x and y variables, respectively, and the slope variable m is calculated as defined in Equation 1.

```
#Age
>x=c(4,4,5,5,7,7,8,9,10,11,12)
#Price
>y=c(6.3,5.8,5.7,4.5,4.5,4.2,4.1,3.1,2.1,2.5,2.2)

#m = Equation(1)
#Numerator of equation 2
>num=sum((x-mean(x))*(y-mean(y)))
#Denominator of equation 2
>den=sum((x-mean(x))^2)
#m=Numerator/Denominator (Equation 1)
>m=num/den
#Equation(3)
>c = mean(y)-m*mean(x)
#Print the slop and interception values
>print(paste("Slop (m)=",round(m,2),"y-intercept (c)= ",c))

[1] "Slop (m)= -0.5 y-intercept (c)= 7.84"
```

This finding may be verified using the R linear regression function `lm()` as given below:

```
#Regression using lm() function
>r=lm(y~x)
> summary(r)

Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-0.8241 -0.1669  0.1807  0.3295  0.4734

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.8363     0.4118  19.027 1.41e-08 ***
x             -0.5024     0.0520  -9.662 4.76e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4614 on 9 degrees of freedom

Multiple R-squared:  0.9121,    Adjusted R-squared:  0.9023
F-statistic: 93.35 on 1 and 9 DF,  p-value: 4.76e-06
```

These values of linear regression of the car's age and price can be further analysed using R plots and an overlaid linear regression slope line using the `abline()` function.

```
> plot(x,y,xlab='Age of Car',ylab='Price of Car',main="Age - Price plot",col='red',cex=2,pch=19)
> abline(r,col="blue",lw=2)
```

The scatter plot and regression line (Figure 2) clearly show a linear relationship between the age and price variables, and the slope indicates a decrease in the car's price with the increase in its age. **END** 🐧

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